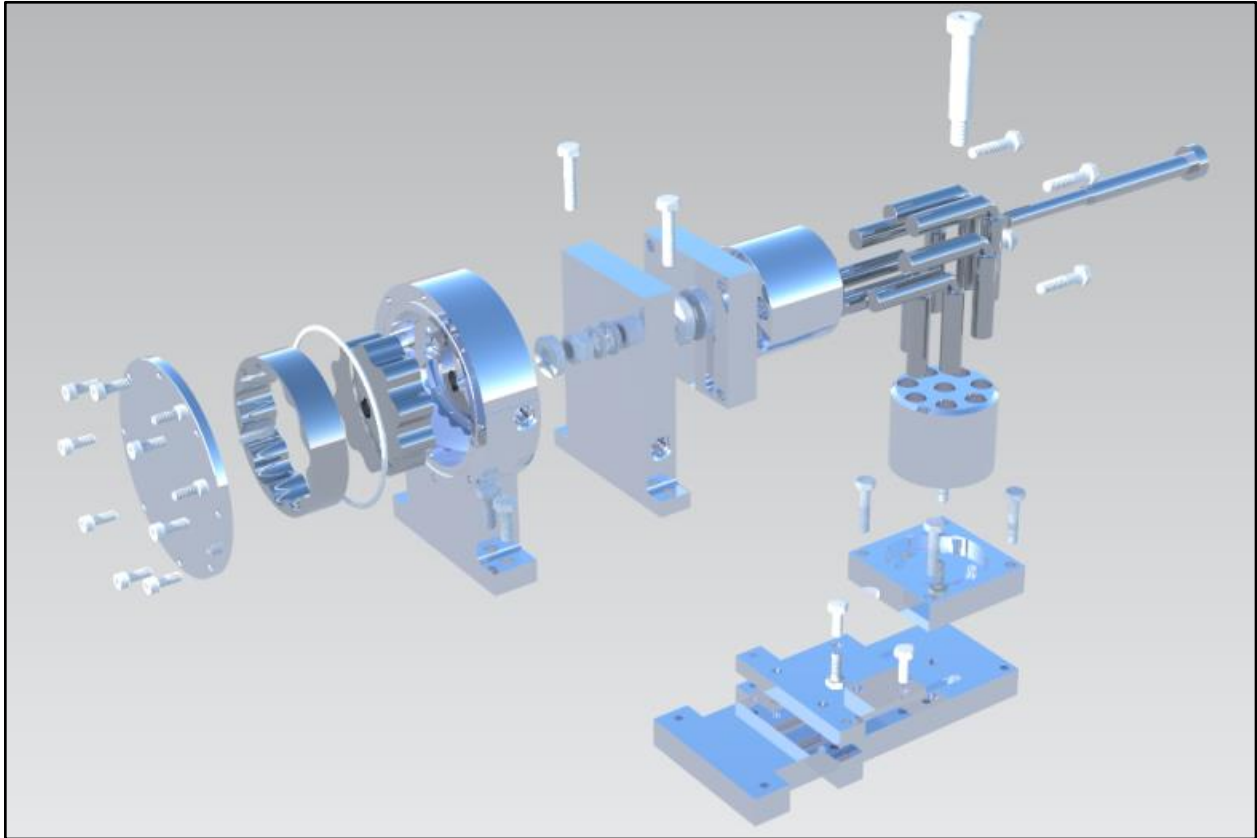


# Lab 04: Gerotor Hydraulic Fluid Pump

Ollie Liang, Jonas Mukund, Yul Kim

ME16400, SEC 011, Spring 2026



## Gerotor Hydraulic Fluid Pump

The Gerotor Hydraulic Fluid Pump (GHFP) is a high-performance positive displacement pump designed to efficiently pressurize fluids through a specialized gear mechanism. As the central shaft rotates, it drives an inner gear that meshes with an outer gear, creating expanding and contracting chambers that draw fluid in and force it out at high pressure. This specific design is housed in a precision-machined base with a universal 3.200-inch internal diameter, allowing for interchangeable gear configurations to suit various hydraulic needs.

To ensure long-term reliability under vibration and thermal stress, the GHFP utilizes a robust mechanical assembly. The shaft is locked axially using a high-strength nylon lock nut and a wedge-locking washer, while torque is transferred via a machine key. The system is sealed by a high-temperature Viton O-ring and a lid secured by eight stainless steel screws, balancing ease of manufacturability with industrial-grade durability for pressurized fluid applications.

| No. | Part Name                         | Modeled / Sourced | Drawings     |
|-----|-----------------------------------|-------------------|--------------|
| 1   | Base                              | Yul Kim           | Jonas Mukund |
| 2   | Main Bearing Support Base         | Jonas Mukund      | Yul Kim      |
| 3   | Piston Housing Base               | Jonas Mukund      | Jonas Mukund |
| 4   | Piston Housing                    | Jonas Mukund      | Yul Kim      |
| 5   | Piston                            | Ollie Liang       | Jonas Mukund |
| 6   | Shaft                             | Ollie Liang /     | Yul Kim      |
| 7   | Piston Housing Valve              | Jonas Mukund      | Jonas Mukund |
| 8   | Main Bearing Support              | Yul Kim           | Jonas Mukund |
| 9   | Housing Base                      | Ollie Liang       | Ollie Liang  |
| 10  | Inner Gerotor                     | Individual        | Individual   |
| 11  | Outer Gerotor                     | Individual        | Individual   |
| 12  | Housing Lid                       | Ollie Liang       | Ollie Liang  |
| 13  | .750 Hex Head Screw               | Ollie Liang       | --           |
| 14  | 1.25 Hex Head Shoulder Screw      | Yul Kim           | --           |
| 15  | 2.625 Alloy Steel Shoulder Screw  | Jonas Mukund      | --           |
| 16  | 0.750 OD Ball Bearing             | Yul Kim           | --           |
| 17  | Coated Steel Wedge Lock Washer    | Ollie Liang       | --           |
| 18  | High-Strength Nylon Lock Nut      | Ollie Liang       | --           |
| 19  | .375 OD Ball Bearing              | Yul Kim           | --           |
| 20  | 3.700 OD Viton O-Ring             | Ollie Liang       | --           |
| 21  | .750 Rounded Machine Key          | Ollie Liang       | --           |
| 22  | .500 Corrosion Resistant SS Screw | Ollie Liang       | --           |

## **Changes and Modifications**

### **Housing Base and Main Bearing Support**

The existing port holes in the housing base were redesigned rather than left as raw openings. Standardized SAE-AS5202 ports were engineered in their place to ensure consistent fluid flow interface and improved sealing compatibility with industry standardized fluid systems.

### **Gerotor Mechanism**

The flywheel, which served as a rotational inertia stabilizer in the original air engine, was removed entirely. In its place, a gerotor gear set was installed to serve as a pumping element. Custom housing was designed and fabricated to tightly enclose the fluid system. This housing connects directly to the base assembly, which also acts as the “shaft support” part at the end of the assembly.

### **Shaft and Fastener Changes**

We replaced the short shaft with a shoulder screw. This provides a controlled and repeatable method of securing the pistons to the base while alleviating manufacturing costs.

We also designed an entirely custom long shaft. The new part interfaces with two ball bearings, a locknut and washer, and modeled holes. The thread ensures proper axial location in the assembly, while ball bearings provide frictionless support.

### **Air Passage and Piston Housing Redesign**

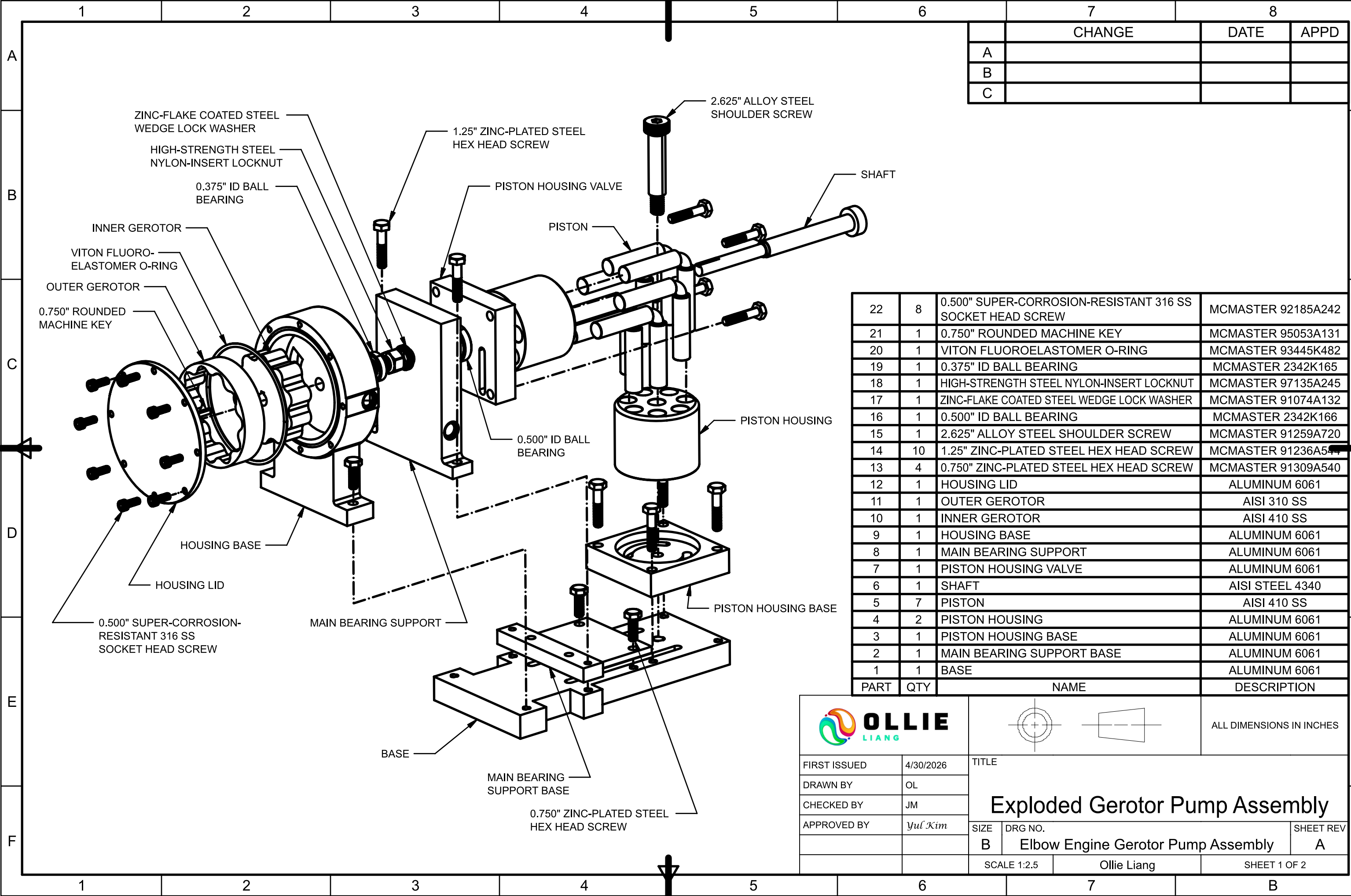
The original design used internal set screw holes and blind bores to manage air routing. These were replaced with surface air passages machined directly into the piston housing base, with other parts acting as the sealing surface. This approach allows for easier inspection and maintenance of fluid passages.

### **Miscellaneous**

The number of cylinders was increased from five to seven.

Custom hardware (screws, O-ring, threaded nut, and ball bearings) was sourced from McMaster-Carr, and custom hardware housing (threaded holes, O-ring grooves, etc.) was also modeled. RC5 fits are applied to all parts where high RPM is to be expected, and FN2 fits are applied between the ball bearings and their housing, and the shaft and upper piston housing to attain a nearly permanent press fit.

Part materials were chosen to maximize ease of manufacturability, while still retaining safety factors at high speeds and temperatures. Importantly, the inner and outer gerotor gears are designed with stainless steel of different hardnesses to prevent galling from occurring.



|   | CHANGE | DATE | APPD |
|---|--------|------|------|
| A |        |      |      |
| B |        |      |      |
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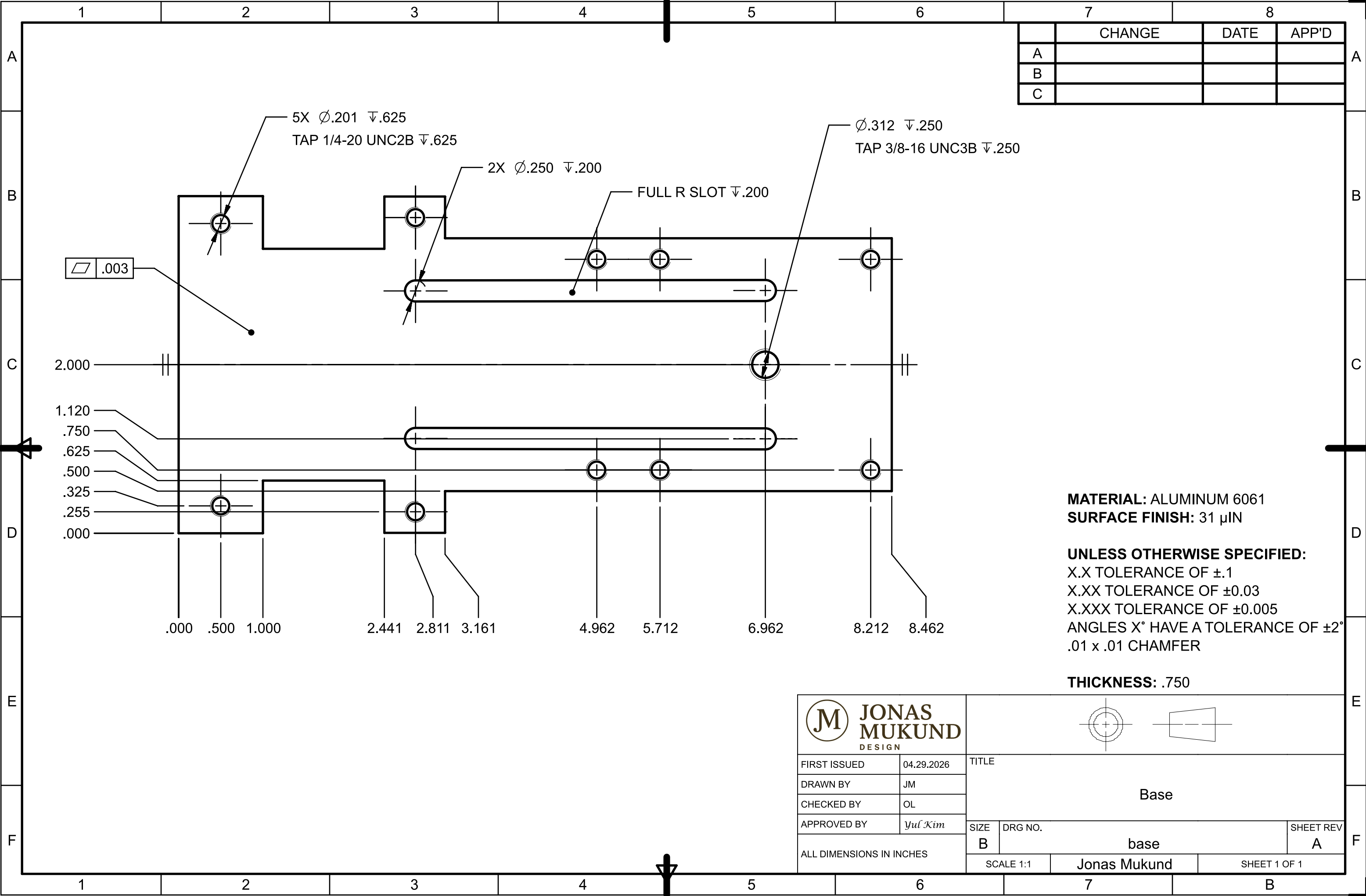
|      |     |                                                           |                    |
|------|-----|-----------------------------------------------------------|--------------------|
| 22   | 8   | 0.500" SUPER-CORROSION-RESISTANT 316 SS SOCKET HEAD SCREW | MCMaster 92185A242 |
| 21   | 1   | 0.750" ROUNDED MACHINE KEY                                | MCMaster 95053A131 |
| 20   | 1   | VITON FLUOROELASTOMER O-RING                              | MCMaster 93445K482 |
| 19   | 1   | 0.375" ID BALL BEARING                                    | MCMaster 2342K165  |
| 18   | 1   | HIGH-STRENGTH STEEL NYLON-INSERT LOCKNUT                  | MCMaster 97135A245 |
| 17   | 1   | ZINC-FLAKE COATED STEEL WEDGE LOCK WASHER                 | MCMaster 91074A132 |
| 16   | 1   | 0.500" ID BALL BEARING                                    | MCMaster 2342K166  |
| 15   | 1   | 2.625" ALLOY STEEL SHOULDER SCREW                         | MCMaster 91259A720 |
| 14   | 10  | 1.25" ZINC-PLATED STEEL HEX HEAD SCREW                    | MCMaster 91236A544 |
| 13   | 4   | 0.750" ZINC-PLATED STEEL HEX HEAD SCREW                   | MCMaster 91309A540 |
| 12   | 1   | HOUSING LID                                               | ALUMINUM 6061      |
| 11   | 1   | OUTER GEROTOR                                             | AISI 310 SS        |
| 10   | 1   | INNER GEROTOR                                             | AISI 410 SS        |
| 9    | 1   | HOUSING BASE                                              | ALUMINUM 6061      |
| 8    | 1   | MAIN BEARING SUPPORT                                      | ALUMINUM 6061      |
| 7    | 1   | PISTON HOUSING VALVE                                      | ALUMINUM 6061      |
| 6    | 1   | SHAFT                                                     | AISI STEEL 4340    |
| 5    | 7   | PISTON                                                    | AISI 410 SS        |
| 4    | 2   | PISTON HOUSING                                            | ALUMINUM 6061      |
| 3    | 1   | PISTON HOUSING BASE                                       | ALUMINUM 6061      |
| 2    | 1   | MAIN BEARING SUPPORT BASE                                 | ALUMINUM 6061      |
| 1    | 1   | BASE                                                      | ALUMINUM 6061      |
| PART | QTY | NAME                                                      | DESCRIPTION        |

|              |           |
|--------------|-----------|
| FIRST ISSUED | 4/30/2026 |
| DRAWN BY     | OL        |
| CHECKED BY   | JM        |
| APPROVED BY  | Yul Kim   |

ALL DIMENSIONS IN INCHES

|                                |                                    |             |
|--------------------------------|------------------------------------|-------------|
| TITLE                          |                                    |             |
| Exploded Gerotor Pump Assembly |                                    |             |
| SIZE                           | DRG NO.                            | SHEET REV   |
| B                              | Elbow Engine Gerotor Pump Assembly | A           |
| SCALE 1:2.5                    |                                    | Ollie Liang |
| SHEET 1 OF 2                   |                                    |             |



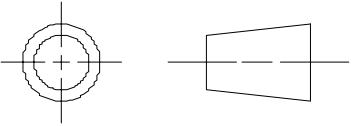


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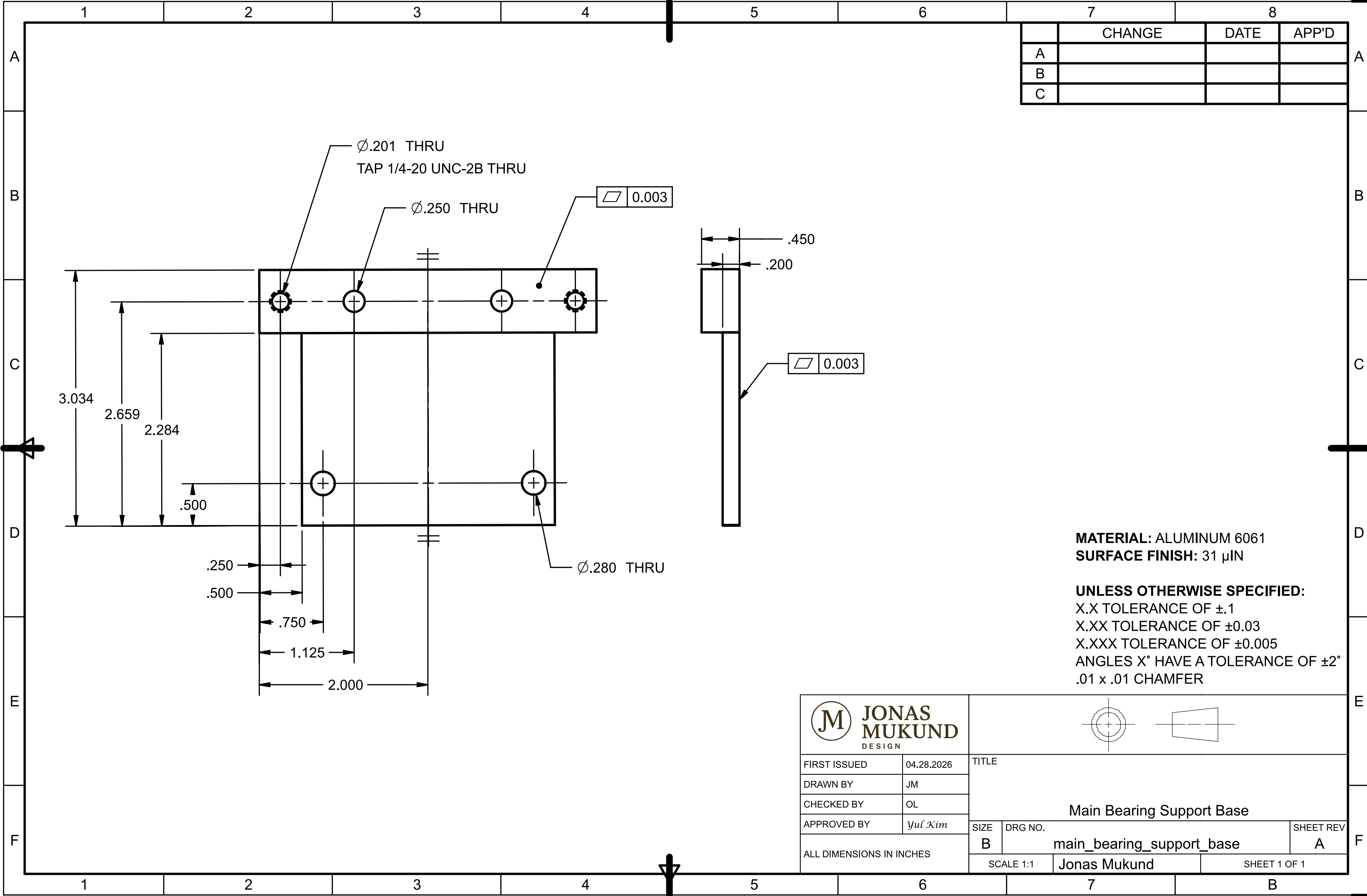


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ALL DIMENSIONS IN INCHES



|           |         |              |      |              |
|-----------|---------|--------------|------|--------------|
| TITLE     |         |              | Base |              |
| SIZE      | DRG NO. | SHEET REV    |      |              |
| B         | base    | A            |      |              |
| SCALE 1:1 |         | Jonas Mukund |      | SHEET 1 OF 1 |



|   | CHANGE | DATE | APP'D |
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| B |        |      |       |
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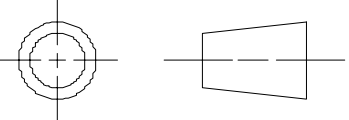
**MATERIAL:** ALUMINUM 6061  
**SURFACE FINISH:** 31 µIN

**UNLESS OTHERWISE SPECIFIED:**  
X.X TOLERANCE OF ±.1  
X.XX TOLERANCE OF ±0.03  
X.XXX TOLERANCE OF ±0.005  
ANGLES X° HAVE A TOLERANCE OF ±2°  
.01 x .01 CHAMFER



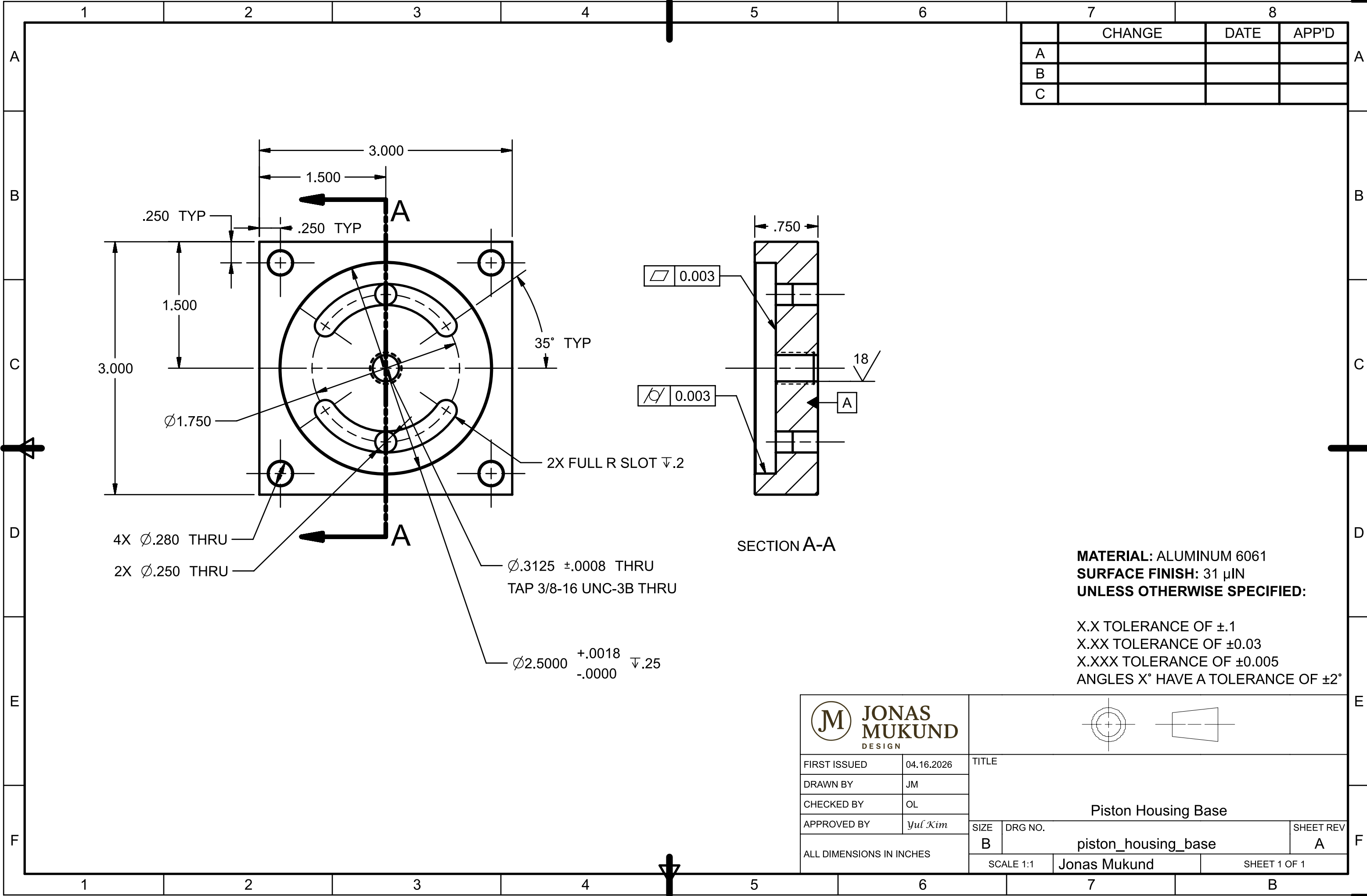
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|--------------|----------------|
| FIRST ISSUED | 04.28.2026     |
| DRAWN BY     | JM             |
| CHECKED BY   | OL             |
| APPROVED BY  | <i>Yul Kim</i> |

ALL DIMENSIONS IN INCHES



|                           |                           |              |  |
|---------------------------|---------------------------|--------------|--|
| TITLE                     |                           |              |  |
| Main Bearing Support Base |                           |              |  |
| SIZE                      | DRG NO.                   | SHEET REV    |  |
| B                         | main_bearing_support_base | A            |  |
| SCALE 1:1                 |                           | SHEET 1 OF 1 |  |





|   | CHANGE | DATE | APP'D |
|---|--------|------|-------|
| A |        |      |       |
| B |        |      |       |
| C |        |      |       |



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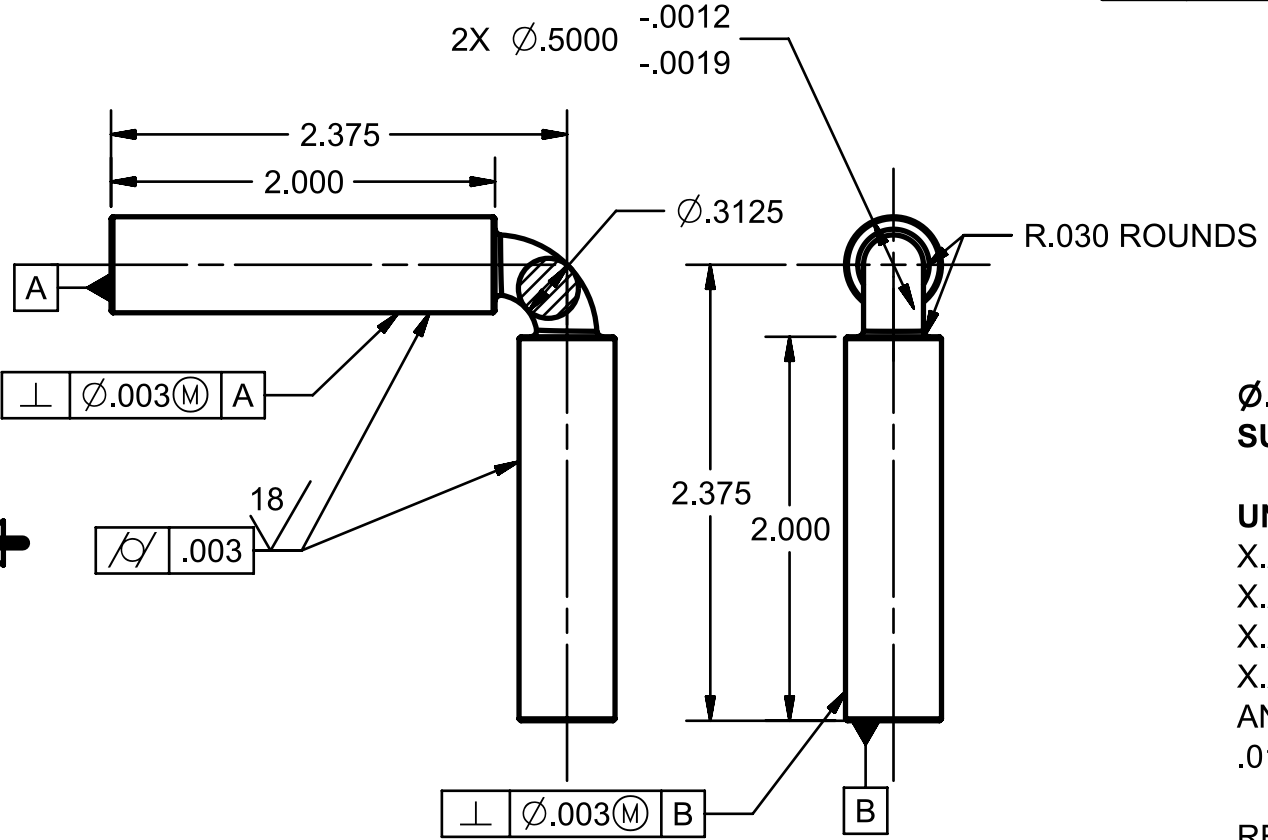
ALL DIMENSIONS IN INCHES

|                     |                     |              |  |
|---------------------|---------------------|--------------|--|
| TITLE               |                     |              |  |
| Piston Housing Base |                     |              |  |
| SIZE                | DRG NO.             | SHEET REV    |  |
| B                   | piston_housing_base | A            |  |
| SCALE 1:1           |                     | SHEET 1 OF 1 |  |





|   | CHANGE | DATE | APP'D |
|---|--------|------|-------|
| A |        |      |       |
| B |        |      |       |
| C |        |      |       |



Ø.MATERIAL: AISI 410 SS  
SURFACE FINISH: 31 µIN

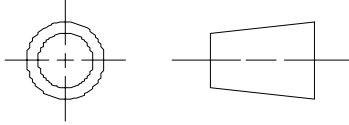
**UNLESS OTHERWISE SPECIFIED:**  
X.X TOLERANCE OF ±.1  
X.XX TOLERANCE OF ±0.03  
X.XXX TOLERANCE OF ±0.005  
X.XXXX TOLERANCE OF ±0.0008  
ANGLES X° HAVE A TOLERANCE OF ±2°  
.01 x .01 CHAMFER

REVOLVED AROUND CENTRAL AXIS

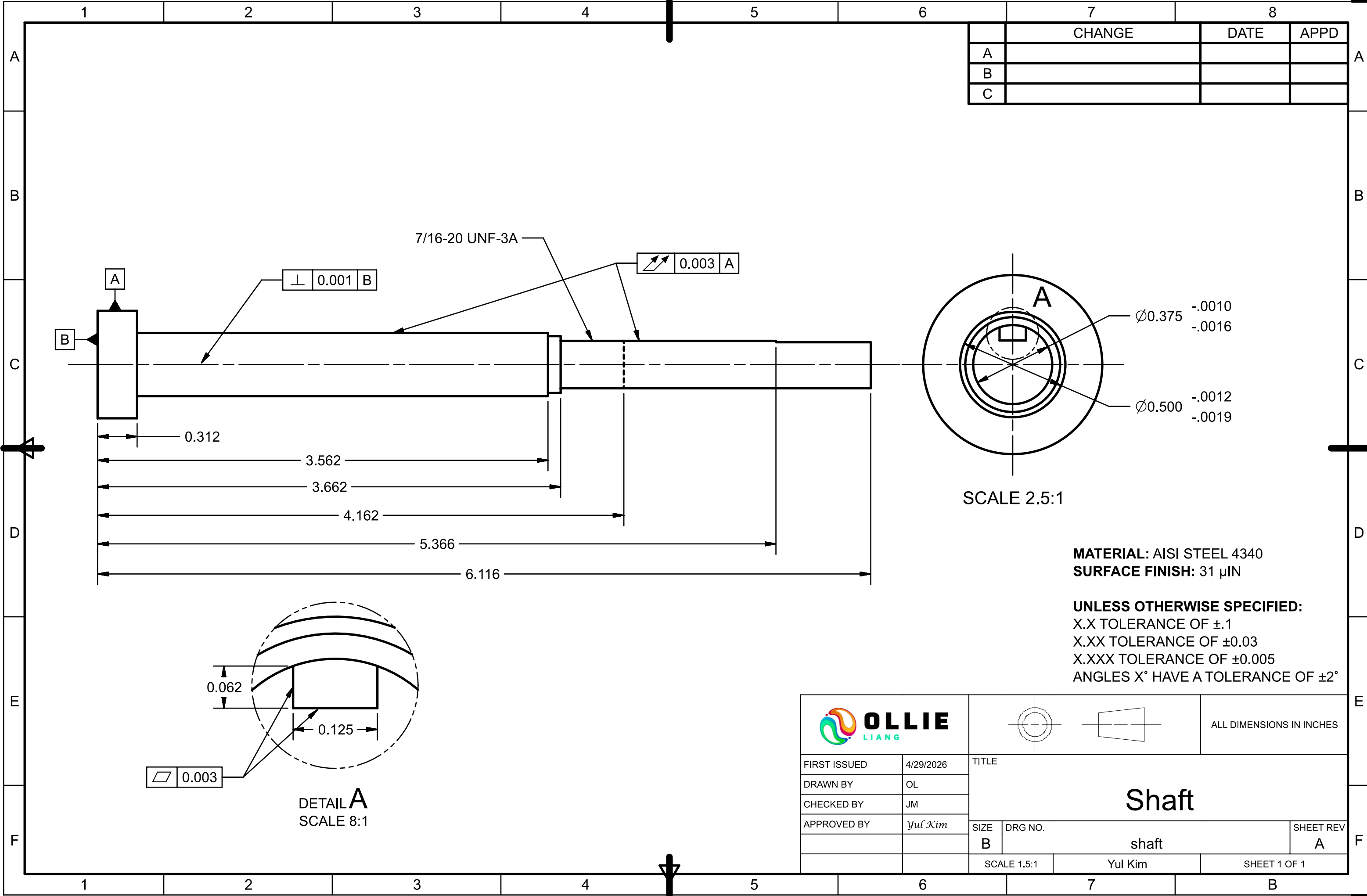


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| CHECKED BY   | OL         |
| APPROVED BY  | yul Kim    |

ALL DIMENSIONS IN INCHES



|           |              |              |
|-----------|--------------|--------------|
| TITLE     |              |              |
| Piston    |              |              |
| SIZE      | DRG NO.      | SHEET REV    |
| A         | piston       | A            |
| SCALE 1:1 | Jonas Mukund | SHEET 2 OF 2 |



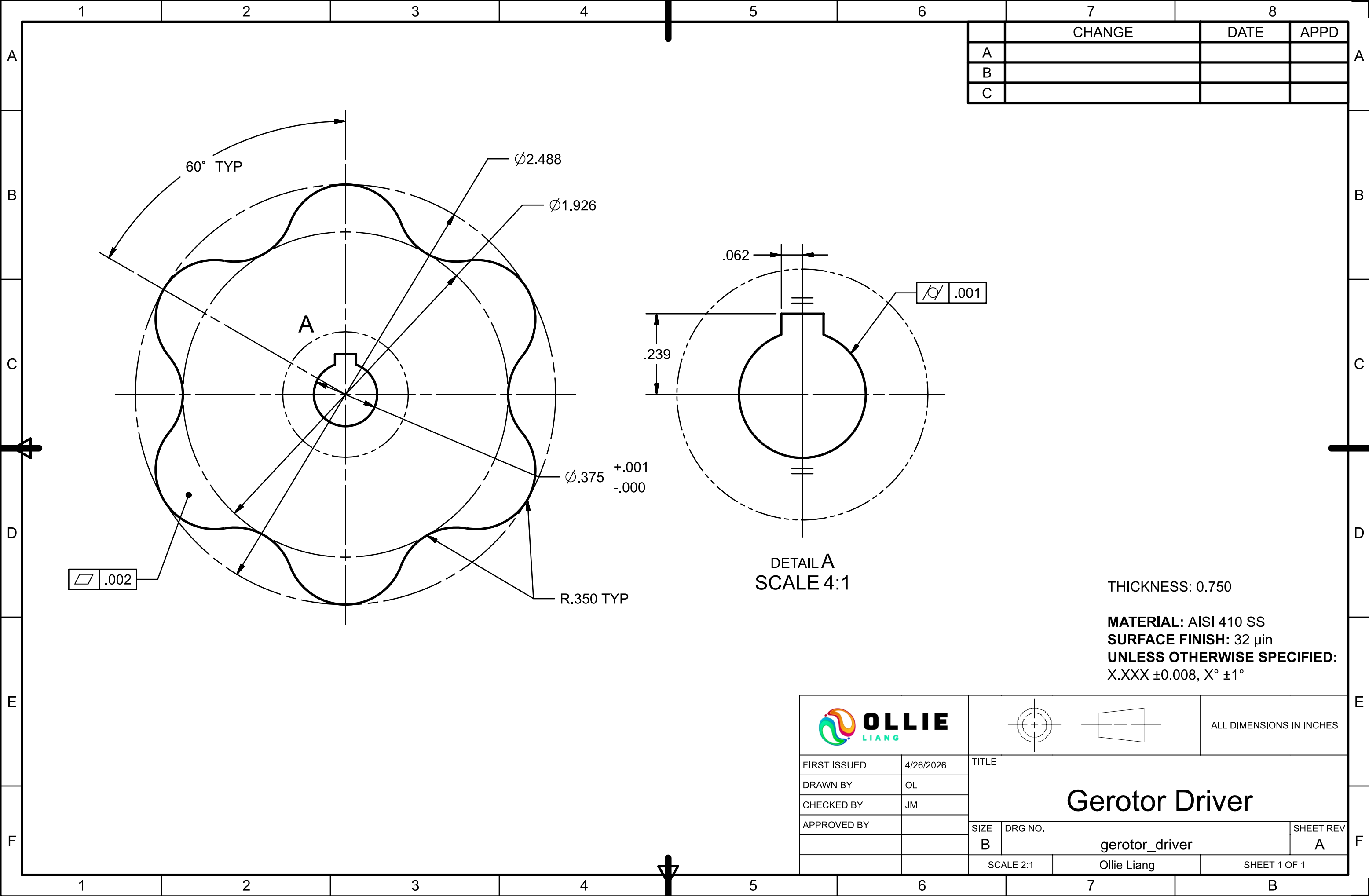




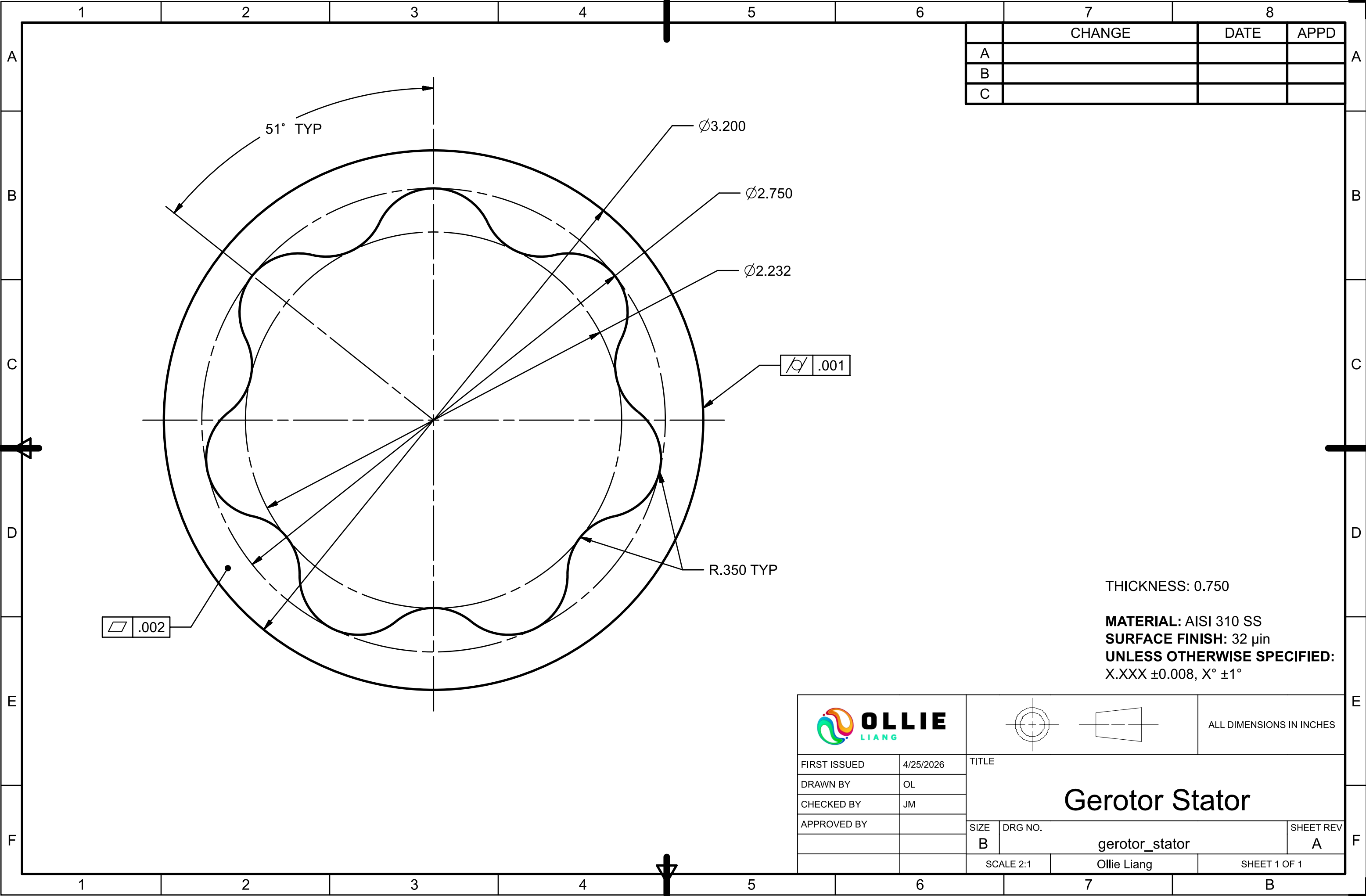








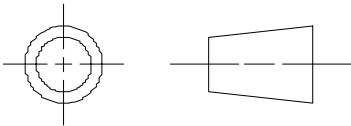




|   | CHANGE | DATE | APPD |
|---|--------|------|------|
| A |        |      |      |
| B |        |      |      |
| C |        |      |      |

THICKNESS: 0.750

**MATERIAL:** AISI 310 SS  
**SURFACE FINISH:** 32 μin  
**UNLESS OTHERWISE SPECIFIED:**  
X.XXX ±0.008, X° ±1°

|                                                                                       |           |                                                                                       |                           |                          |                |
|---------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------|---------------------------|--------------------------|----------------|
|  |           |  |                           | ALL DIMENSIONS IN INCHES |                |
| FIRST ISSUED                                                                          | 4/25/2026 | TITLE<br><br><h1>Gerotor Stator</h1>                                                  |                           |                          |                |
| DRAWN BY                                                                              | OL        |                                                                                       |                           |                          |                |
| CHECKED BY                                                                            | JM        |                                                                                       |                           |                          |                |
| APPROVED BY                                                                           |           |                                                                                       |                           |                          |                |
|                                                                                       |           | SIZE<br>B                                                                             | DRG NO.<br>gerotor_stator |                          | SHEET REV<br>A |
|                                                                                       |           | SCALE 2:1                                                                             |                           | Ollie Liang              | SHEET 1 OF 1   |

